Nick Nguyen

0482814211 | nick.nguyen2150@gmail.com | linkedin.com/in/nicknguyen2150/ | www.nickng.me

Education

The University of Adelaide — Adelaide, SA
Masters of Artificial Intelligence and Machine Learning – GPA: 6.67/7 — *Sep 2025 – Present*

The University of Melbourne — Melbourne, VIC
Bachelor of Science - Computing and Software Systems Major – WAM: 75.6 — *Jul 2022 – Jun 2025*

Experience

Software Developer and Computer Vision Engineer — Oct 2025 – Dec 2025
Legend Sports — *Adelaide, SA*

- Built a spatial sound engine driven by player and ball telemetry, using ffmpeg-python for timestamp alignment, frame demuxing, and dynamic mixing, achieving sub 200 ms sync latency
- Developed a Next.js event-tagging interface with player stats, timeline scrubbing, and hotkeys, reaching annotation throughput of 180 events/hour
- Co-developed a YOLO-based detection model, including Roboflow dataset versioning, augmentations, and hyperparameter sweeps, improving mAP@0.5 by around 10 points

Projects

Basketball Deflection Detection | *PyTorch, Roboflow, YOLOv8, HuggingFace* — Feb 2026 – Jun 2026

- Architected a spatiotemporal computer vision pipeline by integrating a custom-trained object detection model with a Vision-Language Model (SigLIP) for zero-shot team clustering and player identity assignment under single-camera occlusion constraints
- Engineered a temporal sequence classifier using a Bidirectional LSTM (BiLSTM) to process 8-dimensional kinematic feature tensors, implementing Global Motion Estimation (GME) via affine transformations to stabilize auto-panning broadcast footage
- Overcame extreme data scarcity on a constrained 228-clip dataset by implementing F1-optimized early stopping, class-weighted cross-entropy loss, and label smoothing, successfully mitigating overfitting and increasing classification performance from a 0.48 baseline to an F1-score of 0.67

Cybersecurity RAG | *HuggingFace, Streamlit, LangChain, ChromaDB* — April 2026 – Jun 2026

- Developed a Retrieval-Augmented Generation (RAG) pipeline for cybersecurity incident response, enabling secure, zero-cloud-exposure threat triage
- Engineered a Hybrid Search architecture augmented with a Cross-Encoder reranker (BM25), optimizing Top-K retrieval limits and chunk sizes through hyperparameter ablation studies, improving factual faithfulness by 10%
- Benchmarked system across 100+ queries using RAGAS and DeepEval (LLM-as-a-Judge) frameworks, quantitatively evaluating contextual relevancy, information density, and the mitigation of resource bias in generated guidelines

RLHF | *PyTorch, HuggingFace* — Feb 2026 – April 2026

- Engineered a Reinforcement Learning from Human Feedback (RLHF) pipeline to evaluate Proximal Policy Optimization (PPO) against Direct Preference Optimization (DPO) on sub-billion-parameter models (OPT, Pythia)
- Optimized memory bottlenecks for consumer GPU environments by implementing Low-Rank Adaptation (LoRA), 512-token context truncation, and dynamic batch-skipping thresholds to prevent exploding gradients
- Proved DPO's structural stability over PPO by documenting reward-hacking and mode collapse in explicit reward models, demonstrating a $\approx 15\%$ lower perplexity degradation during alignment

Technical Skills

Languages: C, Java, JavaScript, Python, TypeScript, Nix, C#, C++
Developer Tools: Git, Docker, MySQL, Confluence, Trello, Vercel, Figma, Roboflow, VS Code, MATLAB, MongoDB, IntelliJ, SourceTree, Pinecone, ChromaDB, Streamlit, CloudFlare
Libraries: pandas, numPy, matplotlib, tensorflow, scipy, sklearn, ultralytics, torch, scikit-learn, transformers, seaborn, langchain